

Exploiting Out-of-Band Information for Millimeter-Wave MIMO Channel Estimation: Performance in Static and Dynamic Scenarios

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Abstract—To support the high data rates for latency-critical applications, future wireless systems will employ fully digital beamforming multiple-input multiple-output (MIMO) architectures at millimeter wave (mmWave) frequencies. Moreover, mmWave MIMO deployments will coexist with conventional sub-6 GHz MIMO systems, creating opportunities to exploit out-of-band sub-6 GHz information to enhance channel estimation at mmWave frequencies. In this work, we analyze the pilot-aided channel estimation performance of mmWave MIMO systems under various pilot configurations in both static and dynamic environments. We evaluate the system performance in terms of spectral efficiency (SE) for line-of-sight and non-line-of-sight propagation conditions. Simulation results show that incorporating out-of-band sub-6 GHz information yields notable SE gains in both static and dynamic scenarios.

Index Terms—mmWave, digital beamforming, MIMO, channel estimation, out-of-band information.

I. INTRODUCTION

Millimeter wave (mmWave) communication systems are a promising solution for accommodating the rapidly growing demand for high data rates, enabled by the availability of large bandwidths in the mmWave spectrum [1]–[3]. To compensate for the severe path loss inherent at these frequencies, mmWave systems rely on large-scale antenna arrays in conjunction with multiple-input multiple-output (MIMO) techniques [4]. Fully digital beamforming architectures are essential to fully exploit the spatial multiplexing and beamforming capabilities of MIMO systems [5], [6]. Unlike analog and hybrid architectures, digital beamforming enables simultaneous multi-directional channel estimation within a single time interval, thereby significantly reducing link establishment overhead and supporting latency-critical applications [7].

A major challenge in the realization of digital mmWave MIMO systems is reliable link establishment, which depends on accurate channel estimation [4]. At mmWave frequencies, channel estimation is particularly demanding due to the low pre-beamforming signal-to-noise ratio (SNR). In contrast, sub-6 GHz systems experience significantly lower propagation losses, resulting in a higher pre-beamforming SNR [8], [9]. As mmWave systems are increasingly deployed alongside sub-6 GHz systems [10], this reliable out-of-band information can be leveraged to assist and enhance mmWave channel estimation. Several recent studies have investigated the use of

out-of-band information for mmWave channel estimation [11], [12], proposing pilot-aided methods that exploit the line-of-sight (LOS) channel component acquired with the support of the sub-6 GHz band.

The performance of digital mmWave beamforming systems has been studied in static environments [13]–[16], as well as in dynamic scenarios [17]–[19]. However, existing studies on digital mmWave beamforming systems that exploit out-of-band information have largely focused on static environments [12], [20], [21], while dynamic scenarios remain largely unexplored. Moreover, to the best of the author’s knowledge, the impact of different pilot configurations on the performance of digital mmWave MIMO systems in dynamic scenarios has not yet been studied.

Contribution: In this paper, we evaluate the performance of digital beamforming mmWave MIMO systems in both static and dynamic environments. We consider both conventional channel estimation and out-of-band aided channel estimation as proposed in [12]. Our analysis considers various pilot configurations, a wide range of receiver velocities, different SNR levels and both LOS and non-line-of-sight (NLOS) propagation conditions characterized by the Rician K -factor. The performance of the considered methods is assessed through simulations in terms of spectral efficiency (SE).

Organization: Section II describes the system model. Section III outlines the considered channel estimation methods. A simulation-based performance evaluation is presented in Section IV. Finally, Section V concludes the paper.

Notation: The superscript $(\cdot)^{(b)}$ indicates quantities that depend on the operating frequency band, where $b \in \{s, m\}$, with s denoting the sub-6 GHz and m the mmWave band. Scalars are represented by x , vectors by bold lowercase letters $\mathbf{x}$ and matrices by bold uppercase letters $\mathbf{X}$. For a matrix $\mathbf{X}$, $\mathbf{X}_{i,:}$ and $\mathbf{X}_{:,j}$ denote its i -th row and j -th column, respectively. The transpose and Hermitian transpose operators are denoted by $(\cdot)^T$ and $(\cdot)^H$, respectively. The Euclidean norm is written as $\|\cdot\|$, while the Frobenius norm is denoted by $\|\cdot\|_F$.

II. SYSTEM MODEL

We consider a point-to-point multi-band MIMO communication system in which sub-6 GHz and mmWave systems operate simultaneously under far-field propagation conditions.

The transmitter employs $M_{\text{Tx}}^{(s)}$ antenna elements at sub-6 GHz and $M_{\text{Tx}}^{(m)}$ antenna elements at mmWave. Correspondingly, the receiver is equipped with $M_{\text{Rx}}^{(s)}$ sub-6 GHz and $M_{\text{Rx}}^{(m)}$ mmWave antenna elements. Both frequency bands utilize uniform linear arrays (ULAs) composed of dipole antennas, which are approximated as isotropic radiators. The sub-6 GHz and mmWave arrays are assumed to be spatially co-located and aligned, such that the LOS path is characterized by the same angle-of-departure (AoD) ϑ and angle-of-arrival (AoA) φ in both bands [22]. The spacing between adjacent antenna elements in each array is chosen as $\Delta d^{(b)} = 0.5 \lambda^{(b)}$, where $\lambda^{(b)}$ denotes the wavelength of the respective system. We further assume perfect time and frequency synchronization across all transmitting and receiving antennas. The transmitter and receiver are equipped with one radio frequency (RF) chain per antenna, enabling fully digital beamforming in both the sub-6 GHz and mmWave systems. The system follows a time-division duplex (TDD) operation, thereby ensuring reciprocal channel responses.

We consider an orthogonal frequency-division multiplexing (OFDM) transmission system employing $N^{(b)}$ subcarriers and $K^{(b)}$ OFDM time-symbols. For each OFDM time-symbol, a block of $N^{(b)}$ quadrature amplitude modulation (QAM) symbols is assigned to $N^{(b)}$ distinct subcarriers. The OFDM system converts a broadband frequency-selective channel into narrowband frequency-flat channels with the help of a discrete Fourier transform (DFT) and application of a cyclic prefix [23]. The cyclic prefix length is assumed to exceed the maximum excess delay of the multipath channel, thereby eliminating inter-symbol interference. Furthermore, inter-carrier interference, such as that caused by Doppler shifts or carrier frequency offsets, is considered negligible [24].

A. Channel Model

For a given OFDM subcarrier n and time-symbol k , the propagation channel is represented by an $M_{\text{Rx}}^{(b)} \times M_{\text{Tx}}^{(b)}$ dimensional complex-valued channel matrix $\mathbf{H}^{(b)}[n, k]$, based on the equivalent complex baseband model of the OFDM system. These matrices characterize the small-scale fading behavior of the wireless channel incorporating the OFDM processing at the transmitter and receiver. In particular, $\mathbf{H}^{(b)}[n, k]$ corresponds to a sampled version of the underlying channel transfer function (CTF), following the formulation in [25]. In what follows, if the description of a quantity is independent of the subcarrier index n and/or the time-symbol index k , the respective index is omitted for notational simplicity.

We adopt a time-varying frequency-selective channel model with Rician fading, modeled as [26]

$$\mathbf{H}^{(b)}[n, k] = \sqrt{\eta^{(b)}} \sqrt{\frac{\kappa^{(b)}}{1 + \kappa^{(b)}}} \underbrace{\mathbf{H}_{\text{fs}}^{(b)}[k]}_{\text{free-space}} + \sqrt{\eta^{(b)}} \sqrt{\frac{1}{1 + \kappa^{(b)}}} \underbrace{\mathbf{H}_{\text{sp}}^{(b)}[n, k]}_{\text{stochastic part}}, \quad (1)$$

where $\eta^{(b)}$ accounts for large-scale attenuation effects, including path loss and shadowing, while $\kappa^{(b)}$ denotes the Rician K -factor associated with frequency band $b \in \{s, m\}$. To reflect the different propagation characteristics at mmWave and sub-6 GHz frequencies, the mmWave K -factor is assumed to scale with that of the sub-6 GHz band according to $\kappa^{(m)} = c_\kappa \kappa^{(s)}$, where c_κ denotes the scaling factor. The deterministic LOS channel component corresponds to a free-space propagation model and is defined by

$$\mathbf{H}_{\text{fs}}^{(b)}[k] = e^{j\chi^{(b)}} e^{j \frac{2\pi \nu^{(b)} k}{\Delta f^{(b)}}} \mathbf{a}_{\text{Rx}}^{(b)}(\varphi) \left(\mathbf{a}_{\text{Tx}}^{(b)}(\vartheta) \right)^H \quad (2)$$

where

$$\mathbf{a}_{\text{Tx}}^{(b)}(\vartheta) = \begin{bmatrix} 1 & \dots & e^{-j2\pi \left(M_{\text{Tx}}^{(b)} - 1 \right) \frac{\Delta d^{(b)}}{\lambda^{(b)}} \sin(\vartheta)} \end{bmatrix}^T \quad (3)$$

and

$$\mathbf{a}_{\text{Rx}}^{(b)}(\varphi) = \begin{bmatrix} 1 & \dots & e^{-j2\pi \left(M_{\text{Rx}}^{(b)} - 1 \right) \frac{\Delta d^{(b)}}{\lambda^{(b)}} \sin(\varphi)} \end{bmatrix}^T \quad (4)$$

denote the array steering vectors associated with the AoD ϑ and AoA φ , respectively. In (2), the term $\chi^{(b)} \sim \mathcal{U}(-\pi, \pi)$ models the random phase offset at the first (reference) antenna element, $\nu^{(b)}$ denotes the Doppler shift and $\Delta f^{(b)}$ represents the subcarrier spacing. The stochastic channel matrix $\mathbf{H}_{\text{sp}}^{(b)}[n, k]$ is generated according to the 3rd Generation Partnership Project (3GPP) channel model [27]. Given the large difference between the sub-6 GHz and mmWave frequencies, the stochastic channels $\mathbf{H}_{\text{sp}}^{(s)}[n, k]$ and $\mathbf{H}_{\text{sp}}^{(m)}[n, k]$ are assumed to be statistically independent. Furthermore, a channel model following Jakes sum-of-sinusoids approach [28] is employed to determine the temporal evolution of the wireless channel [29].

B. Link Establishment

The establishment of a communication link consists of a training phase, followed by a data transmission phase. During the training phase, the wireless channel is estimated independently at sub-6 GHz and mmWave frequency bands. The channel estimates from both bands are then utilized jointly for data transmission at the mmWave band.

1) *Training Phase:* In the training phase, known pilot symbols are transmitted from each antenna element and are assumed to be perfectly known at the receiver. The pilot symbols $\phi^{(b)}[n, k] \in \mathbb{C}^{M_{\text{Tx}}^{(b)} \times 1}$ generated from the QAM constellation are distributed across $N^{(b)}$ subcarriers such that each transmit antenna is assigned a distinct subset of non-overlapping subcarriers.

For the pilot allocation, we consider three pilot patterns similar to those defined by the 3GPP for the channel state information reference signal (CSI-RS) [30]. These pilot patterns are designed to facilitate efficient and scalable channel estimation for different antenna configurations and deployment scenarios. Note that the pilot patterns considered in this work are provided as illustrative examples and are not fully compliant with the 3GPP standard. These pilot patterns are

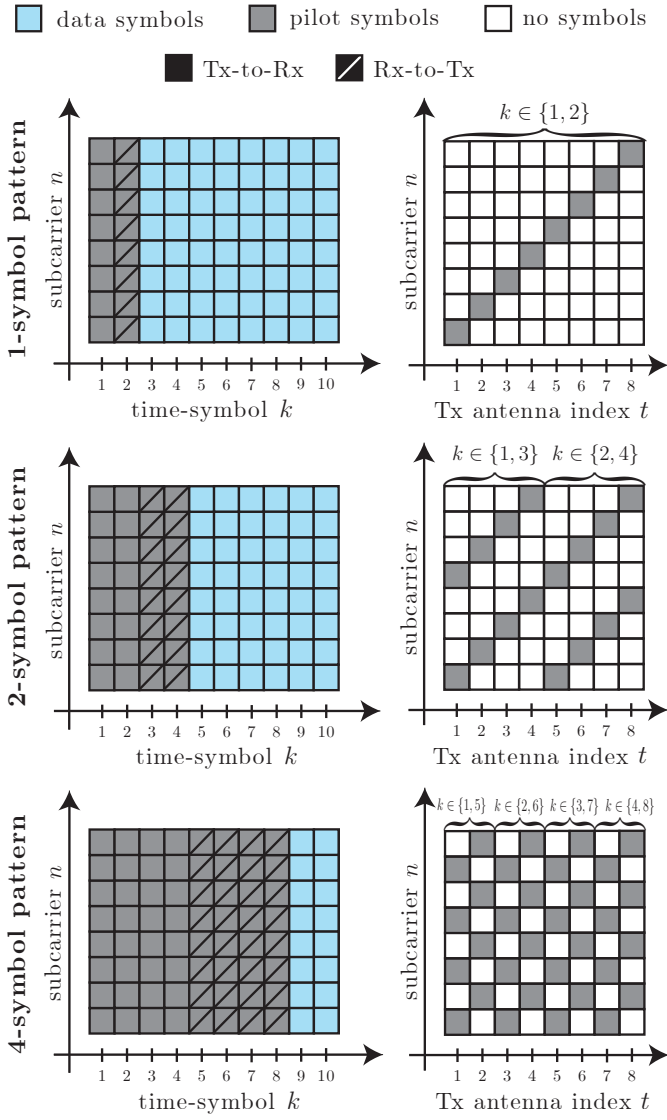


Fig. 1. Pilot symbols at each antenna occupy a certain number of subcarriers such that they do not overlap in the frequency domain. The 1-symbol pattern (top) confines all pilot transmissions to a single OFDM time-symbol, the 2-symbol pattern (middle) expands the pilot allocation over two OFDM time-symbols, while the 4-symbol pattern (bottom) spans four OFDM time-symbols per transmission direction.

intended for potential use in future systems and may be adapted based on the specific configuration and deployment scenario. In this work, we refer to them colloquially as the 1-symbol, 2-symbol, and 4-symbol pilot patterns, based on the number $K_P^{(b)}$ of OFDM time-symbols they span. Note that the system operates using the TDD protocol, which requires the pilot symbols to be allocated for both the transmitter-to-receiver (Tx-to-Rx) and receiver-to-transmitter (Rx-to-Tx) directions. Since we typically use the same number of transmit and receive antennas in this paper, we set the number of time-symbols $K_P^{(b)}$ to be identical for both the Tx-to-Rx and Rx-to-Tx directions. This constraint can, however, be relaxed when the numbers of transmit and receive antennas differ. An

example pilot allocation for a system with $M_{Tx}^{(b)} = 8$ transmit antennas is illustrated in Fig. 1. The pilot allocation at the t -th transmit antenna is given by

$$\phi_t^{(b)}[n, k] = \begin{cases} \phi^{(b)}[n, k], & \text{if } (n, k, t) \in \mathcal{P}^{(b)} \\ 0, & \text{else} \end{cases} \quad (5)$$

where the set of pilot positions $\mathcal{P}^{(b)}$ is defined by

$$\begin{aligned} \mathcal{P}^{(b)} = \{ (n, k, t) : \\ n \in \left\{ t_m, t_m + \frac{M_{Tx}^{(b)}}{K_P^{(b)}}, \dots, N^{(b)} - \frac{M_{Tx}^{(b)}}{K_P^{(b)}} + t_m \right\}, \\ k \in \{1, 2, \dots, 2K_P^{(b)}\}, \quad t \in \{1, 2, \dots, M_{Tx}^{(b)}\}, \\ t_m = \left(t - 1 \bmod \frac{M_{Tx}^{(b)}}{K_P^{(b)}} \right) + 1 \}. \end{aligned} \quad (6)$$

and $t \in \{1, \dots, M_{Tx}^{(b)}\}$ denotes the transmit antenna index.

The 1-symbol pilot pattern ($K_P^{(b)} = 1$) confines all pilot transmissions to a single OFDM time-symbol per transmission direction. This enables fully parallel pilot transmission and offers minimal time-domain overhead, making it attractive for latency-critical applications. However, the limited number of pilot resource elements restricts its applicability to systems with few transmit antennas and may result in reduced estimation accuracy in frequency-selective channels due to sparse pilot density in the frequency domain.

The 2-symbol pilot pattern ($K_P^{(b)} = 2$) expands the pilot allocation over two OFDM time-symbols per transmission direction. This extension increases the available pilot resources, supporting a larger number of antennas and improving frequency-domain resolution. This denser pilot placement leads to more reliable channel estimation in frequency-selective environments, at the expense of higher time-domain overhead compared to the 1-symbol pattern.

In the 4-symbol pilot pattern ($K_P^{(b)} = 4$), pilots span four OFDM time-symbols, providing high pilot density and robust channel estimation, particularly for large-scale MIMO systems. The improved accuracy comes with a significant increase in overhead, which reduces resources available for data transmission and increases latency.

The input-output relationship of the training phase is given by

$$\mathbf{y}^{(b)}[n, k] = \mathbf{H}^{(b)}[n, k] \phi^{(b)}[n, k] + \mathbf{w}^{(b)}[n, k], \quad (7)$$

where the received signal is denoted by $\mathbf{y}^{(b)}[n, k] \in \mathbb{C}^{M_{Rx}^{(b)} \times 1}$ and the additive white Gaussian noise (AWGN) with the power of $\sigma_{w^{(b)}}^2$ is denoted by $\mathbf{w}^{(b)}[n, k] \sim \mathcal{CN}(0, \sigma_{w^{(b)}}^2 \mathbf{I}_{M_{Rx}^{(b)}})$. At the receiver, least-squares (LS) channel estimates are first obtained at the pilot subcarriers. The channel response at the remaining subcarriers is then obtained via linear interpolation, yielding the estimated channel matrix $\tilde{\mathbf{H}}^{(b)}[n] \in \mathbb{C}^{M_{Rx}^{(b)} \times M_{Tx}^{(b)}}$. Since the channel estimation spans over $K_P^{(b)}$ symbols to obtain a full MIMO channel estimate, we consider the estimated

quantities as time-invariant and therefore omit the time-index k . However, it is important to note that the actual channel $\mathbf{H}^{(b)}[n, k]$ may still vary over time.

Since channel estimation is performed at the receiver, the corresponding channel state information (CSI) has to be available also at the transmitter to enable optimal beamforming. Owing to the use of a TDD scheme, the transmitter acquires the CSI by leveraging channel reciprocity in both frequency bands. Specifically, the procedure begins with the transmission of training symbols from the transmitter to the receiver (Tx-to-Rx), allowing the receiver to acquire the CSI. This is followed by the transmission of training symbols in the reverse direction (Rx-to-Tx), enabling the transmitter to acquire the CSI.

2) *Data Transmission*: During the data transmission phase, only the mmWave link is considered. Data symbols are transmitted over $N^{(m)}$ subcarriers and $K_D^{(m)} = K^{(m)} - K_P^{(m)}$ OFDM time-symbols. The channel estimates obtained in the training phase are further processed using the methods which will be detailed in Section III. The resulting mmWave channel estimate $\bar{\mathbf{H}}^{(m)}[n]$ is then employed to design the precoding and combining matrices. To maximize SE, singular value decomposition (SVD) is adopted. The compact-form SVD of $\bar{\mathbf{H}}^{(m)}[n]$ is given by

$$\bar{\mathbf{H}}^{(m)}[n] = \bar{\mathbf{Q}}^{(m)}[n] \bar{\mathbf{\Sigma}}^{(m)}[n] \left(\bar{\mathbf{F}}^{(m)}[n] \right)^H, \quad (8)$$

where $\bar{\mathbf{Q}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(m)} \times \ell_{\text{max}}}$ contains the left singular vectors and serves as the combining matrix, $\bar{\mathbf{F}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Tx}}^{(m)} \times \ell_{\text{max}}}$ comprises the right singular vectors and defines the precoder, $\bar{\mathbf{\Sigma}}^{(m)}[n]$ denotes the diagonal matrix of singular values $\text{diag}(\bar{\sigma}_{(1)}^{(m)}[n], \dots, \bar{\sigma}_{(\ell_{\text{max}})}^{(m)}[n])$ and $\ell_{\text{max}} = \min(M_{\text{Rx}}^{(m)}, M_{\text{Tx}}^{(m)})$ denotes the maximum number of spatial streams. The power loading matrix is defined by $\bar{\mathbf{P}}^{(m)}[n] = \text{diag}(\bar{p}_{(1)}^{(m)}[n], \dots, \bar{p}_{(\ell_{\text{max}})}^{(m)}[n])$ and is optimized via the water-filling algorithm to maximize the achievable rate under a total transmit power constraint

$$\left\| \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n] \right)^{1/2} \right\|_F^2 = P_T^{(m)}, \quad (9)$$

where $P_T^{(m)}$ denotes the total transmit power. With this notation, the input-output relationship for the data transmission phase is then given by

$$\mathbf{y}^{(m)}[n, k] = \left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{H}^{(m)}[n, k] \bar{\mathbf{x}}^{(m)}[n, k] + \left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{w}^{(m)}[n, k], \quad (10)$$

where the precoded transmit signal is denoted by $\bar{\mathbf{x}}^{(m)}[n, k] = \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n] \right)^{1/2} \mathbf{x}^{(m)}[n, k]$, the unprecoded transmit signal is denoted by $\mathbf{x}^{(m)}[n, k] \in \mathbb{C}^{\ell_{\text{max}} \times 1}$, the received signal is denoted by $\mathbf{y}^{(m)}[n, k] \in \mathbb{C}^{\ell_{\text{max}} \times 1}$ and the AWGN added at the mmWave receiver is denoted by

TABLE I
SIMULATION PARAMETERS

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
Number of Data Symbols K_D	N/A	7
Transmit Power P_T	30 dBm	30 dBm
Antenna Configuration ($M_{\text{Rx}} \times M_{\text{Tx}}$)	8×8	8×8
RMS Delay Spread σ_τ	1148 ns	841 ns
Number of Realizations L_r	1000	1000

$\mathbf{w}^{(m)}[n, k] \sim \mathcal{CN}(0, \sigma_{w^{(m)}}^2 \mathbf{I}_{M_{\text{Rx}}^{(m)}})$. Since $\bar{\mathbf{Q}}^{(m)}[n]$ is semi-unitary, $\left(\bar{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{w}^{(m)}[n, k]$ has the same statistical distribution as $\mathbf{w}^{(m)}[n, k]$.

III. CHANNEL ESTIMATION METHODS

This section outlines the mmWave channel estimation schemes considered in this work.

A. Conventional

The conventional approach uses only the in-band mmWave channel estimate $\tilde{\mathbf{H}}^{(m)}[n]$, obtained via LS estimation at pilot subcarriers and subsequent linear interpolation, as described in Section II-B1. The resulting mmWave channel estimate is given by

$$\bar{\mathbf{H}}^{(m)}[n] = \tilde{\mathbf{H}}^{(m)}[n]. \quad (11)$$

Note that this approach does not leverage any additional side information or advanced signal processing techniques.

B. Out-of-band Aided MRC (OOBA-MRC)

This method applies out-of-band aided maximal ratio combining (OOBA-MRC) to optimally combine the in-band channel estimate $\tilde{\mathbf{H}}^{(m)}[n]$ (detailed in Section II-B1) with the out-of-band aided channel estimate $\hat{\mathbf{H}}^{(m)}[n]$ (introduced in [12]). The combined mmWave channel estimate is given by

$$\bar{\mathbf{H}}^{(m)}[n] = \hat{w} \hat{\mathbf{H}}^{(m)}[n] + (1 - \hat{w}) \tilde{\mathbf{H}}^{(m)}[n] \quad (12)$$

where

$$\hat{w} \approx \frac{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)} \sigma_{w^{(m)}}^2}{\frac{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}}{1 + \tilde{\kappa}^{(s)}} + \left(1 + M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)} \right) \sigma_{w^{(m)}}^2}. \quad (13)$$

denotes the approximated optimal combining factor, whose derivation is detailed in [12]. In (13), $\tilde{\kappa}^{(s)}$ denotes the Rician K -factor in the sub-6 GHz band, obtained via the method of moments [31]. This method is designed to adapt to dynamic wireless channel conditions with regard to the Rician K -factor and the noise variance $\sigma_{w^{(m)}}^2$.

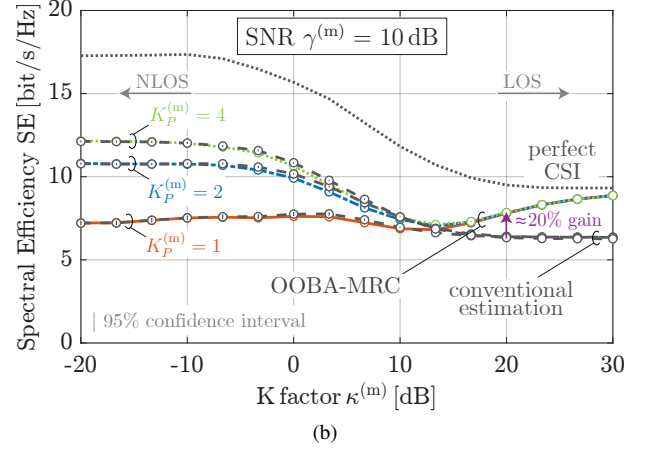
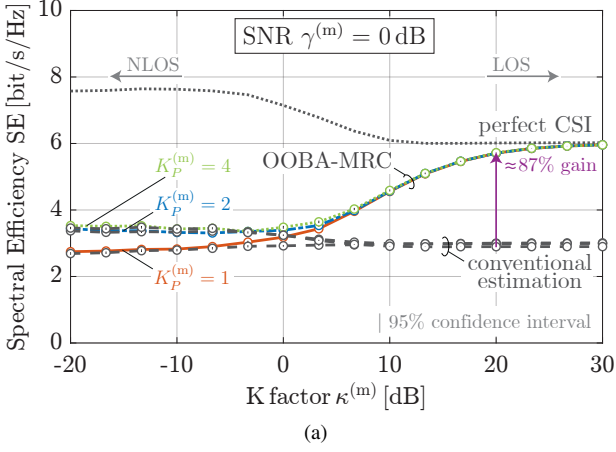


Fig. 2. At a lower SNR of 0 dB, increasing the number of pilot symbols does not yield a significant improvement in SE (left), whereas at a higher SNR of 10 dB, the performance gains become more pronounced (right).

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the channel estimation methods introduced in Section III and the impact of different pilot schemes using Monte-Carlo simulations. We consider the achievable SE as the main performance metric for MIMO systems. Throughout the analysis, we assume equal power allocation across all subcarriers. For the frequency-selective channel model, we employ the 3GPP model for the urban macro LOS scenario with a configurable K -factor, as specified in [27]. Additionally, we assume that the AoD and AoA are mutually independent and uniformly distributed within the range $[-90^\circ, 90^\circ]$. Given the close similarity between sub-6 GHz and mmWave K -factors observed in the measured scenario reported in [32], we set the scaling factor $c_\kappa = 1$ ($c_\kappa = 0$ dB), thereby defining the mmWave K -factor as $\kappa^{(m)} = \kappa^{(s)}$. The corresponding simulation parameters are summarized in Tab. I, which also includes the parameters for the sub-6 GHz MIMO system required for the Ooba-MRC method. In addition to the methods described in Section III, we also include the performance for the case of perfect CSI $\bar{\mathbf{H}}^{(m)}[n] = \mathbf{H}^{(m)}[n]$ to establish an upper performance bound.

The achievable SE in bits/s/Hz averaged over $N^{(m)}$ subcarriers and $K_D^{(m)}$ time-symbols is given by

$$\text{SE} = \frac{1}{N^{(m)}K_D^{(m)}} \sum_{n=1}^{N^{(m)}} \sum_{k=K_P^{(m)}+1}^{K^{(m)}} \sum_{\mu=1}^{\ell_{\max}} \log_2(1 + \text{SINR}_\mu[n, k]) \quad (14)$$

with the effective signal-to-interference-and-noise ratio (SINR) for the stream μ denoted by [33]

$$\text{SINR}_\mu[n, k] = \frac{|\bar{\mathbf{G}}_{\mu, \mu}^{(m)}[n, k]|^2}{\sum_{\substack{\nu=1 \\ \nu \neq \mu}}^{\ell_{\max}} |\bar{\mathbf{G}}_{\mu, \nu}^{(m)}[n, k]|^2 + \sigma^2 \|\bar{\mathbf{Q}}_{:, \mu}^{(m)}[n, k]\|^2}. \quad (15)$$

In (15), the element $\bar{\mathbf{G}}_{\mu, \nu}^{(m)}[n, k]$, with $\mu, \nu \in \{1, \dots, \ell_{\max}\}$, denotes an entry of the channel gain matrix $\bar{\mathbf{G}}^{(m)}[n, k] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ for the n -th subcarrier and k -th time-symbol. The matrix is defined as

$$\bar{\mathbf{G}}^{(m)}[n, k] = \left(\bar{\mathbf{Q}}^{(m)}[n]\right)^H \mathbf{H}^{(m)}[n, k] \bar{\mathbf{F}}^{(m)}[n] \left(\bar{\mathbf{P}}^{(m)}[n]\right)^{1/2}. \quad (16)$$

The overall error variance is given by $\sigma^2 = \sigma_\mu^2[n, k] + \sigma_{w^{(m)}}^2$, where $\sigma_\mu^2[n, k]$ corresponds to the diagonal elements of the estimation error covariance matrix $\bar{\mathbf{C}}_\varepsilon^{(m)}[n, k] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$. The covariance matrix is computed as

$$\bar{\mathbf{C}}_\varepsilon^{(m)}[n, k] = \frac{1}{\ell_{\max}} \bar{\varepsilon}^{(m)}[n, k] \left(\bar{\varepsilon}^{(m)}[n, k]\right)^H, \quad (17)$$

where the estimation error matrix is defined by

$$\bar{\varepsilon}^{(m)}[n, k] = \mathbf{G}^{(m)}[n, k] - \bar{\mathbf{G}}^{(m)}[n, k]. \quad (18)$$

In (18), $\bar{\varepsilon}^{(m)}[n, k]$ quantifies the error between the channel gain matrix obtained using the estimated precoder and combiner, $\bar{\mathbf{G}}^{(m)}[n, k]$, and that obtained with the perfectly designed precoder and combiner, $\mathbf{G}^{(m)}[n, k]$.

A. Performance in Static Scenarios

In this subsection, we investigate the impact of different pilot schemes (see Section II-B1) on the considered channel estimation methods as a function of the K -factor in static scenarios. The simulation results in terms of achievable SE are shown in Fig. 2.

The Ooba-MRC method achieves equal or better performance than the conventional method, regardless of the pilot pattern. For K -factors below 0 dB, increasing the number of pilot symbols generally improves the SE, since more pilots yield more accurate channel estimates. As the K -factor increases beyond 0 dB, the performance obtained with different pilot patterns converges, because the channel becomes increasingly dominated by the LOS component and adding more pilot symbols in the frequency domain does not

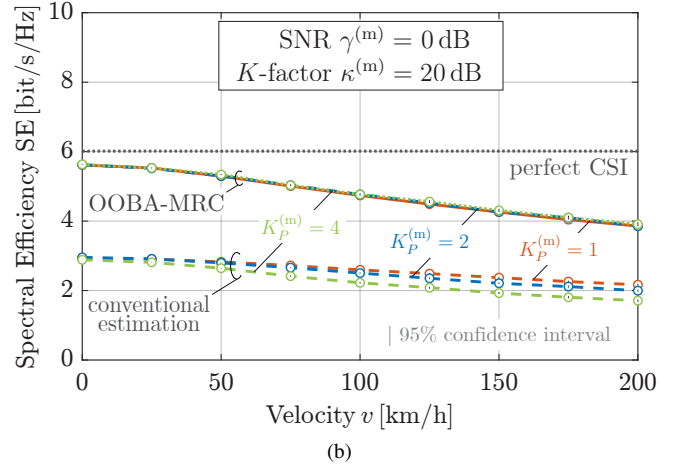
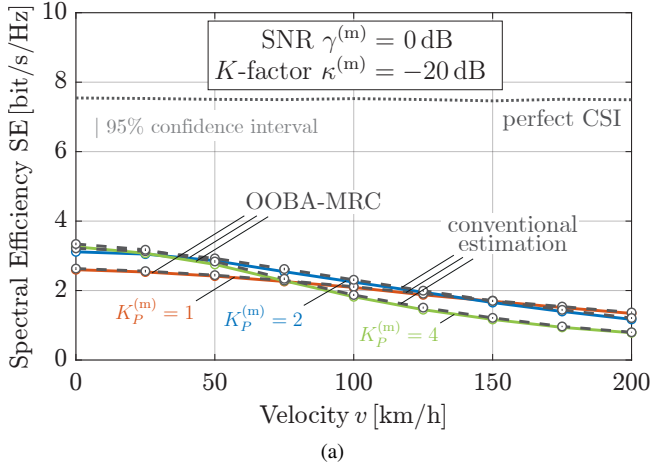


Fig. 3. At a low K -factor of -20 dB, the Ooba-MRC method provides no gain in achievable SE compared to the conventional method (left) and at a high K -factor, its relative gain increases with the number of pilot symbols (right).

provide additional benefit. For an SNR of 0 dB and low K -factor, the SE improves from 2.7 bit/s/Hz (1-symbol pattern) to 3.4 bit/s/Hz (2-symbol pattern), and only slightly further to 3.5 bit/s/Hz (4-symbol pattern). For an SNR of 10 dB, the gains from additional pilot symbols are more pronounced: the SE increases from 7 bit/s/Hz (1-symbol pattern) to 11 bit/s/Hz (2-symbol pattern) and to 12.5 bit/s/Hz (4-symbol pattern).

B. Performance in Dynamic Scenarios

In this subsection, we analyze the performance of the considered estimation methods under time-varying channel conditions as a function of the receiver velocity. Simulation results for an mmWave SNR of 0 dB in terms of achievable SE are shown in Fig. 3. Both NLOS and LOS propagation conditions are considered, characterized by K -factors of -20 dB (see Fig. 3a) and 20 dB (see Fig. 3b), respectively.

Two counteracting effects are of particular interest here. Increasing the number of pilot symbols enhances the estimation accuracy by providing more reliable channel information. However, increasing the number of pilot symbols causes channel aging, where the estimates lose relevance as the channel changes over time. Thus, it is important to evaluate the resulting trade-off between estimation accuracy and channel aging. As expected, the achievable SE generally decreases with increasing velocity, due to faster channel variations, regardless of the K -factor, estimation method or pilot pattern employed.

In the NLOS scenario with a K -factor of -20 dB, the Ooba-MRC method primarily relies on the in-band channel estimate, resulting in performance that closely matches that of the conventional method, regardless of the pilot pattern. For low-velocities (up to approximately 50 km/h), the SE increases with the number of pilot symbols $K_P^{(m)}$, as the enhanced estimation accuracy dominates. In the time-invariant case, the 1-symbol pilot pattern yields an SE of 2.7 bit/s/Hz, the 2-symbol pattern increases the SE to 3.4 bit/s/Hz and the 4-symbol pattern reaches 3.5 bit/s/Hz. With increasing velocity, however, channel aging becomes dominant, leading to a faster

SE degradation for larger pilot patterns. At 200 km/h, the 1-symbol pattern achieves the highest SE of 1.3 bit/s/Hz, while the 2-symbol pattern slightly decreases to 1.2 bit/s/Hz and the 4-symbol pattern performs worst at 0.8 bit/s/Hz. The 1-symbol pattern begins to outperform the 4-symbol pattern at approximately 75 km/h and surpasses the 2-symbol pattern from about 150 km/h onward.

In the LOS scenario with a K -factor of 20 dB, the Ooba-MRC method outperforms the conventional method by dominantly exploiting the out-of-band aided LOS channel estimate. Its performance remains essentially independent of the number of pilot symbols. In contrast, the conventional method is affected by channel aging, causing performance to deteriorate as the number of pilot symbols increases. When the receiver velocity reaches 200 km/h, the SE drops from 2.9 bit/s/Hz in the time-invariant case to 2.2 bit/s/Hz for the 1-symbol pattern, 2 bit/s/Hz for the 2-symbol pattern and 1.7 bit/s/Hz for the 4-symbol pattern. For the Ooba-MRC method, varying the number of pilot symbols has no noticeable impact, with the SE decreasing from 5.6 bit/s/Hz in the static case to 3.9 bit/s/Hz at 200 km/h.

V. CONCLUSION

In this paper, we investigate the channel estimation performance of digital beamforming mmWave MIMO systems and the impact of different pilot schemes in both static and dynamic scenarios. Simulation results confirm the effectiveness of the Ooba-MRC method compared to the conventional method. The SNR and Rician K -factor affect the performance of the Ooba-MRC method to a great extent.

In static scenarios, increasing the number of pilot symbols improves the quality of the channel estimates. For low K -factors, where the channel is highly frequency-selective, this results in an overall increase in SE. However, for high K -factors where the channel is predominantly flat, increasing the number of pilot symbols does not improve the SE performance.

In dynamic scenarios, channel aging becomes a dominant factor. As the number of pilot symbols increases, the channel may change significantly during the estimation interval, causing older estimates to lose relevance. Consequently, for low K -factors, lower velocities benefit from additional pilots, leading to increased SE, while higher velocities experience a decline in SE as more pilots are used. For high K -factors, the conventional method exhibits decreasing SE with an increasing number of pilots, whereas Ooba-MRC remains robust and maintains high performance even at high mobility.

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